What was simulated

1. Every flat pattern folded in 3D with its bend table (direction, angle, inside radius, K = 0.46) - self-clash check and formed envelope vs the source drawing.
2. Assembly stack-up: do the formed parts add up to the 1741.09 overall width and stack as a box.
3. Press-brake: tonnage per bend (304, 8T V-die), shortest flange vs die opening, radii below 1T.
4. Structural hand checks: recovery lug, side hoop, centre section as a beam, chassis bracket fasteners.
5. Nesting on 2500 × 1250 sheets per thickness.

Verdict

FOLDS	All 13 flats fold without self-clash. Formed sections match the source drawing within 1 mm (bracket 115.1 vs 116, P2 238.1 vs 238.64, P3 82.7 vs 83, upright joggle 56.3 vs 56.32, centre Z 80.0 vs 80.34) - the residual is the K-factor difference between the source model and K = 0.46.
STACK-UP	835 + 2×5 + 448.28 + 447.81 = 1741.09 - exact match to the overall width. Parts fit.
BRAKE	1 bend(s) have a flange shorter than the 8T die allows - swap to the V listed in the brake table (gooseneck punch for the centre panel Z). Max tonnage 22 t on the 835 mm bends - needs a 40 t+ brake.
STRUCTURE	Centre section OK for a 10 kN push. Side hoops are cosmetic/handhold grade. The 8 mm lug is NOT a recovery point as drawn - tie-down only. Bracket slots suit M8 - upsize to M12 for a 3 t vehicle.
NESTING	2 mm: 1 sheet(s) at 2%, 3 mm: 1 sheet(s) at 35%, 5 mm: 1 sheet(s) at 10%, 8 mm: 1 sheet(s) at 1%

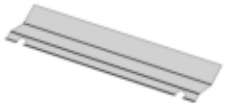
Bottom line

It will build. The flats are consistent with the drawing and with each other. Before you cut steel: (a) fix the two tooling items in the brake table, (b) decide whether the lugs are recovery points - if yes, go to 10-12 mm and R35, (c) open the bracket slots to M12. Tube lengths are drawing-derived - prove the first hoop on the jig.

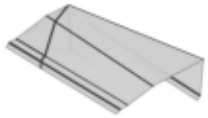
Part	Name	Fold	Envelope (mm)	Formed section W x H (dwg)	Drawing reference
VA-BB-01	Centre panel	OK	835 × 185 × 69	165.7 × 80.0 (dwg 80.34 × 82.15)	sheet 2: section 82.15 / 80.34 / 48.26 / 15 lip,
VA-BB-02	End panel (item 2)	OK	626 × 514 × 269		
VA-BB-03	Upright RH	OK	306 × 763 × 43	759.9 × 56.3 (dwg 56.32)	sheet 4: 56.32 joggle offset, 744.48 tall
VA-BB-04	End panel, opposite hand (item	OK	626 × 514 × 269		
VA-BB-05	End plate	OK	264 × 78 × 14	263.7 × 19.3	sheet 6: 78 × 176.58 plan, 25.47 flange at 40.84
VA-BB-06	Bracket 1 (chassis bracket)	OK	184 × 213 × 55	115.1 × 78.6 (dwg 116 × 79)	sheet 7: 183.63 × 116.00 × 79.00 U-channel
VA-BB-07	Recovery lug	flat part	-		
VA-BB-11	Spot bracket (light mount)	OK	60 × 77 × 17	73.1 × 21.0	sheet 10: 60 wide saddle
VA-BB-12	Centre panel P2 (top plate)	OK	835 × 261 × 33	238.1 × 39.7 (dwg 238.64 × 40)	sheet 11: 238.64 wide, legs 40 / 20, 835 long
VA-BB-13	Centre panel P3 (lower channel	OK	835 × 115 × 34	82.7 × 39.8 (dwg 83 × 40)	sheet 12: 83 × 40 channel, 835 long
VA-BB-14	Upright LH	OK	306 × 763 × 43	759.9 × 56.3 (dwg 56.32)	
VA-BB-17	Face 1 (RH trim strip)	OK	652 × 87 × 214	531.0 x 248.5	
VA-BB-18	Face 1 LH (trim strip)	OK	652 × 87 × 214	531.0 x 248.5	

Fold engine: polygon split at each bend line, bend zone tessellated into 6 sub-folds across the bend allowance so the neutral axis length is preserved; UP rotates toward the sheet normal of the datum face. Envelope is the bounding box of the folded sheet surface in the datum-face frame (compare to outside dims less one thickness where applicable).

VA-BB-01 Centre panel



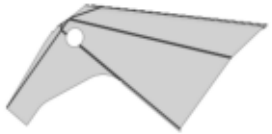
VA-BB-02 End panel (item 2)



VA-BB-03 Upright RH



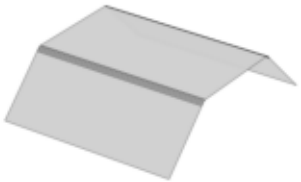
VA-BB-04 End panel, opposite hand (item 4)



VA-BB-05 End plate



VA-BB-06 Bracket 1 (chassis bracket)



VA-BB-11 Spot bracket (light mount)



VA-BB-12 Centre panel P2 (top plate)



VA-BB-13 Centre panel P3 (lower channel)



VA-BB-14 Upright LH



VA-BB-17 Face 1 (RH trim strip)



VA-BB-18 Face 1 LH (trim strip)



Part	Bend	Dir	Angle	R	t	Length	V-die	Tonnes	Min flange	Verdict
VA-BB-01	B1	UP	90°	2.5	3	835	24	22.1	12.5	SHORT FLANGE 12.5 < 17.0 → use V=18 | R<1T sharp-nose
VA-BB-01	B2	DOWN	54.76°	2.5	3	835	24	22.1	33.0	OK | R<1T sharp-nose
VA-BB-01	B3	UP	54.76°	2.5	3	835	24	22.1	79.5	OK | R<1T sharp-nose
VA-BB-02	B1	DOWN	90°	2.5	3	485	24	12.8	-	OK | R<1T sharp-nose
VA-BB-02	B2	DOWN	48°	2.5	3	499	24	13.2	-	OK | R<1T sharp-nose
VA-BB-02	B3	UP	45°	2.5	3	503	24	13.3	-	OK | R<1T sharp-nose
VA-BB-02	B4	DOWN	90°	2.5	3	504	24	13.3	-	OK | R<1T sharp-nose
VA-BB-02	B5	DOWN	90°	2.5	3	296	24	7.8	-	OK | R<1T sharp-nose
VA-BB-02	B6	DOWN	90°	2.5	3	216	24	5.7	-	OK | R<1T sharp-nose
VA-BB-03	B1	DOWN	20°	3	5	186	40	8.2	150.0	OK | R<1T sharp-nose
VA-BB-03	B2	UP	20°	3	5	141	40	6.2	150.0	OK | R<1T sharp-nose
VA-BB-04	B1	DOWN	90°	2.5	3	485	24	12.8	-	OK | R<1T sharp-nose
VA-BB-04	B2	DOWN	48°	2.5	3	499	24	13.2	-	OK | R<1T sharp-nose
VA-BB-04	B3	UP	45°	2.5	3	503	24	13.3	-	OK | R<1T sharp-nose
VA-BB-04	B4	DOWN	90°	2.5	3	504	24	13.3	-	OK | R<1T sharp-nose
VA-BB-04	B5	DOWN	90°	2.5	3	296	24	7.8	-	OK | R<1T sharp-nose
VA-BB-04	B6	DOWN	90°	2.5	3	216	24	5.7	-	OK | R<1T sharp-nose
VA-BB-05	B1	UP	40.84°	1.5	3	78	24	2.1	25.5	OK | R<1T sharp-nose
VA-BB-06	B1	DOWN	90°	3	5	184	40	8.1	74.9	OK | R<1T sharp-nose
VA-BB-06	B2	DOWN	90°	3	5	184	40	8.1	74.9	OK | R<1T sharp-nose
VA-BB-11	B1	DOWN	50°	1.25	2	50	16	0.9	25.2	OK | R<1T sharp-nose
VA-BB-11	B2	DOWN	50°	1.25	2	50	16	0.9	25.2	OK | R<1T sharp-nose
VA-BB-12	B1	UP	90°	2	3	835	24	22.1	17.5	OK | R<1T sharp-nose
VA-BB-12	B2	UP	90°	2	3	835	24	22.1	37.5	OK | R<1T sharp-nose
VA-BB-13	B1	UP	90°	2.5	3	835	24	22.1	37.5	OK | R<1T sharp-nose
VA-BB-13	B2	UP	90°	2.5	3	835	24	22.1	37.5	OK | R<1T sharp-nose
VA-BB-14	B1	DOWN	20°	3	5	186	40	8.2	150.0	OK | R<1T sharp-nose
VA-BB-14	B2	UP	20°	3	5	141	40	6.2	150.0	OK | R<1T sharp-nose
VA-BB-17	B1	DOWN	90°	1.25	2	35	16	0.6	247.0	OK | R<1T sharp-nose
VA-BB-17	B2	DOWN	17°	1.25	2	35	16	0.6	84.4	OK | R<1T sharp-nose
VA-BB-17	B3	DOWN	162°	42	2	35	-	-	-	roll/bump form (R42)
VA-BB-18	B1	DOWN	90°	1.25	2	35	16	0.6	247.0	OK | R<1T sharp-nose
VA-BB-18	B2	DOWN	17°	1.25	2	35	16	0.6	84.4	OK | R<1T sharp-nose
VA-BB-18	B3	DOWN	162°	42	2	35	-	-	-	roll/bump form (R42)

Tonnage: F = 1.33 x UTS x L x t^2 / V (air bend), UTS 520 MPa. Min flange = 0.71 x V for bends >= 60 deg (0.43 x V for shallow bends). R < 1T flagged for 304 - sharp-nose punch, prove on first-off. 162 deg R42 is a roll / bump-form, not a V-die bend.

Recovery lug VA-BB-07 (8 mm 304, dia 30, R30 boss)

Net tension at hole: 2 × 15 × 8 × 210 MPa = 50 kN (5.1 t) at yield

Shear tear-out (2 planes x 15 mm): 30 kN (3.1 t) at yield

Pin bearing (dia 30 × 8, 1.5 Fy): 76 kN (7.7 t)

Governing: tear-out 30 kN (3.1 t) at yield, ~75 kN (7.6 t) ultimate per lug - before the weld to the 3 mm end panel is considered.

VERDICT: fine as a tie-down / mounting lug. NOT a rated snatch-recovery point for a 2.5-3 t vehicle (want >= 2 x GVM with a safety factor, and the load path welded through to the chassis bracket): go to 10-12 mm, R35 boss, and weld to the bracket, or label it tie-down only.

Side hoop VA-BB-15/16 (CHS 50 x 2.0, 285 mm above end panel)

I = 87010 mm4, Z = 3480 mm3

Lateral load at the top to reach yield at the root weld: 2.56 kN (262 kgf)

VERDICT: adequate for brush/animal contact and as a handhold; it is not a structural member. Root weld to the 3 mm end panel is the weak link - add a 3 mm doubler or weld through to the end plate.

Centre section as a beam, 5 kN frontal push at mid-span, 700 mm between chassis brackets

I (sum of centre Z + P2 hat + P3 channel, no composite credit) = 726.5 cm4; Z ~ 161.4 cm3

M = 875 Nm → sigma = 5 MPa vs 210 MPa yield → OK (margin 39x; conclusion holds even if the section is 10x less stiff than summed)

Centre section as a beam, 10 kN frontal push at mid-span, 700 mm between chassis brackets

I (sum of centre Z + P2 hat + P3 channel, no composite credit) = 726.5 cm4; Z ~ 161.4 cm3

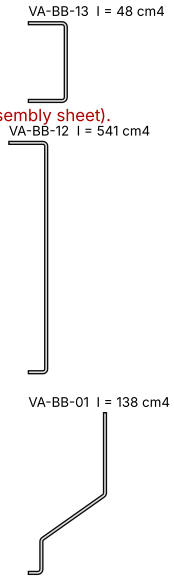
M = 1750 Nm → sigma = 11 MPa vs 210 MPa yield → OK (margin 19x; conclusion holds even if the section is 10x less stiff than summed)

Centre section as a beam, 20 kN frontal push at mid-span, 700 mm between chassis brackets

I (sum of centre Z + P2 hat + P3 channel, no composite credit) = 726.5 cm4; Z ~ 161.4 cm3

M = 3500 Nm → sigma = 22 MPa vs 210 MPa yield → OK (margin 10x; conclusion holds even if the section is 10x less stiff than summed)

Formed sections used for the beam check



Chassis brackets VA-BB-06 (5 mm U, 4 slots each)

Slot 22 × 9 (source) suits M8. For a bull bar on a 3 t vehicle use M12 8.8 minimum: open slots to 13 wide / 26 long on the next revision.

Bracket web 116 × 5 mm in bending under 10 kN at 80 mm lever: sigma ~ 10000 × 80 / (116 × 25 / 6) = 1655 MPa → relies on the centre channels, NOT the bracket alone. Weld brackets to both P2 and P3 (notes on assembly sheet).

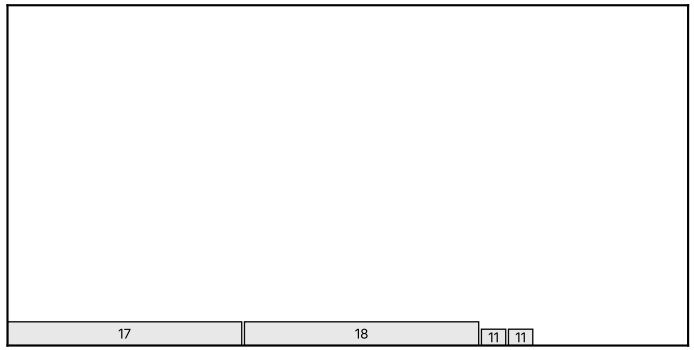
Assembly stack-up (from flats)

835 (centre) + 2 × 5 (uprights) + 448.28 + 447.81 (ends) = 1741.09 = overall width on sheet 1. Fit is exact.

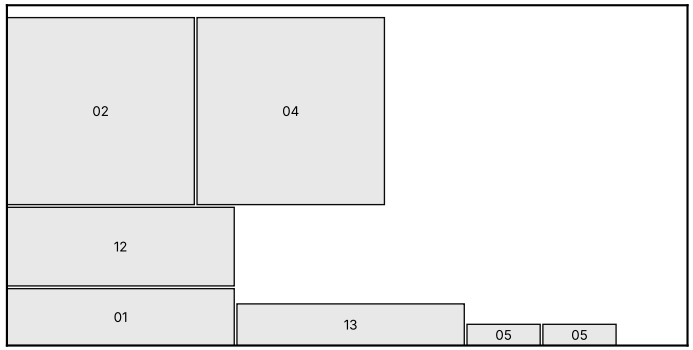
P2 hat (238.64 formed) / P3 channel (83 formed) / centre Z (80.34 deep) all 835 long - they stack as one box; no interference.

Side hoop end: 204 mm inboard leg lands inside the upright window (window 108 × 141) - clearance 50 mm tube in 108 window OK.

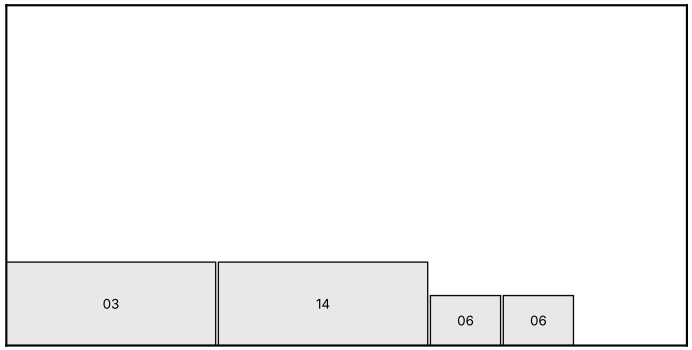
2 mm sheet 1/1 (2% net util. over all sheets)



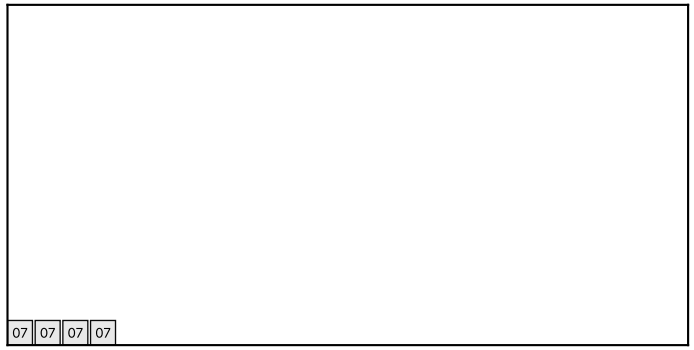
3 mm sheet 1/1 (35% net util. over all sheets)



5 mm sheet 1/1 (10% net util. over all sheets)



8 mm sheet 1/1 (1% net util. over all sheets)



Bounding-box shelf nest for quoting only - a CAM nester with true-shape nesting and common-line cutting will do better, especially on the 3 mm sheet where the two end panels (688 × 687) dominate.